Caesar To Vigenère: Evolution Of Substitution Ciphers

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Introduction to Substitution Ciphers

Definition: A substitution cipher is a method of hiding a message by replacing each letter of the original text with another letter or symbol, using a fixed rule called a key.

- Substitution ciphers hide a message by swapping its letters for other letters or symbols.

Example: With a shift of 3, A becomes D, B becomes E, C becomes F.

- These ciphers have grown from simple to complex over time.

Simple (Caesar): Every letter shifts by the same amount.

Complex (Vigenère): A keyword gives each letter a different shift. With the key KEY:

- Learning this history helps us understand modern security.

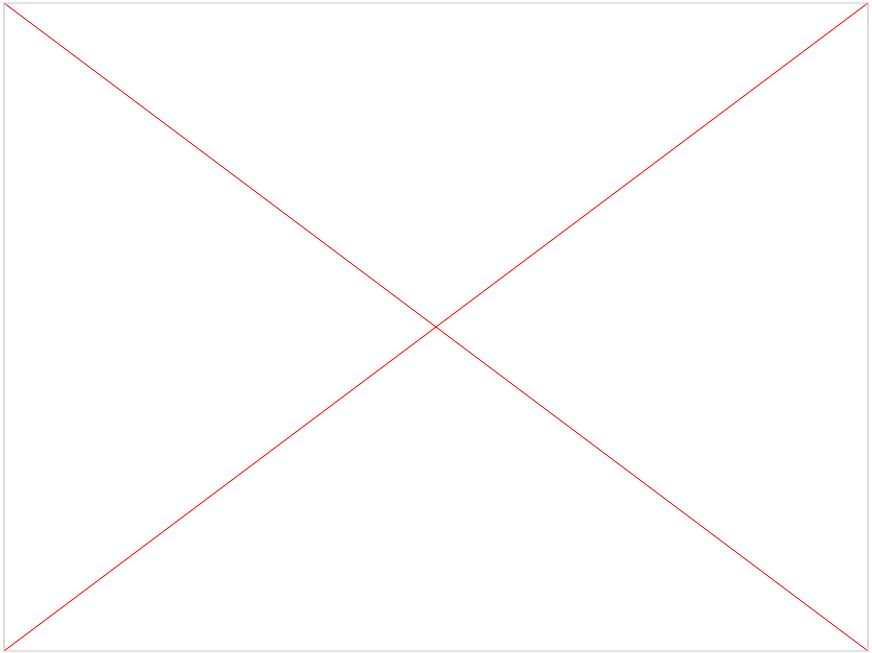
Example: The ideas of a key and encryption from the Caesar cipher are used today in messaging apps like WhatsApp. The key is just much longer and the math is much stronger.

The Caesar Cipher

Caesar cipher shifts each letter by a fixed number. Example: HELLO → KHOOR

It is simple and one of the earliest ciphers. Example: shift 3 forward to encrypt, 3 back to decrypt

It is easy to break. Example: only 25 shifts to try or check the most frequent letter



Advancements in Substitution Ciphers

- Polyalphabetic ciphers use many alphabets. Example: Vigenère
- Homophonic ciphers give one letter many symbols. Example: E → 5, 17, 42
- Polygraphic ciphers encrypt letter groups. Example: Playfair uses pairs
- Each advancement was broken in time, leading to modern ciphers like AES

The Vigenère Cipher

- Vigenère uses a keyword to apply many Caesar shifts. Example: key KEY → shifts 10, 4, 24
- Each letter is shifted differently, which hides frequency patterns. Example: ATTACK → KXRKGI
- It can still be broken, but it was a major step in cipher evolution. Example: Kasiski examination finds the key length

Impact on Modern Cryptography

Early ciphers gave us the ideas of keys and substitution. Example: Caesar’s shift is the ancestor of the AES key

Modern ciphers combine substitution and transposition. Example: AES uses S-boxes and row shifting over many rounds

Learning Caesar to Vigenère shows why strong security is needed. Example: every broken cipher led to a stronger one

